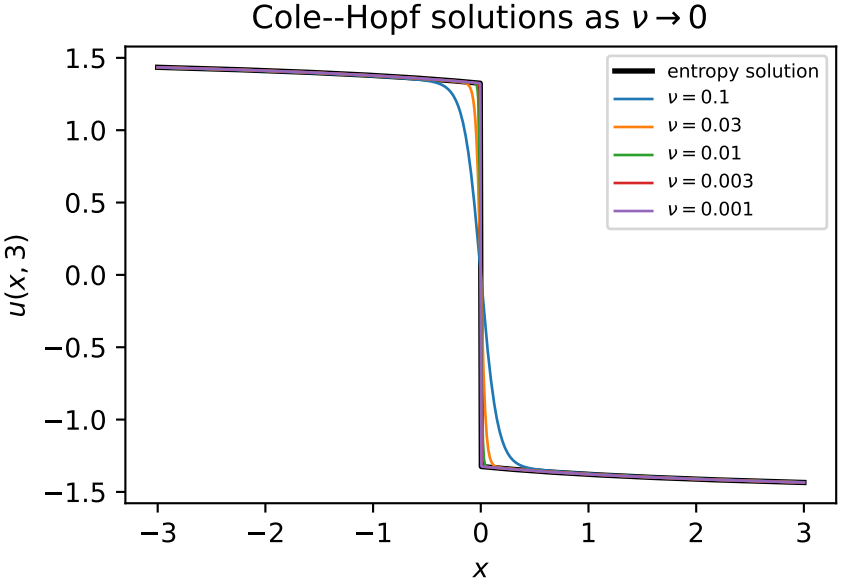
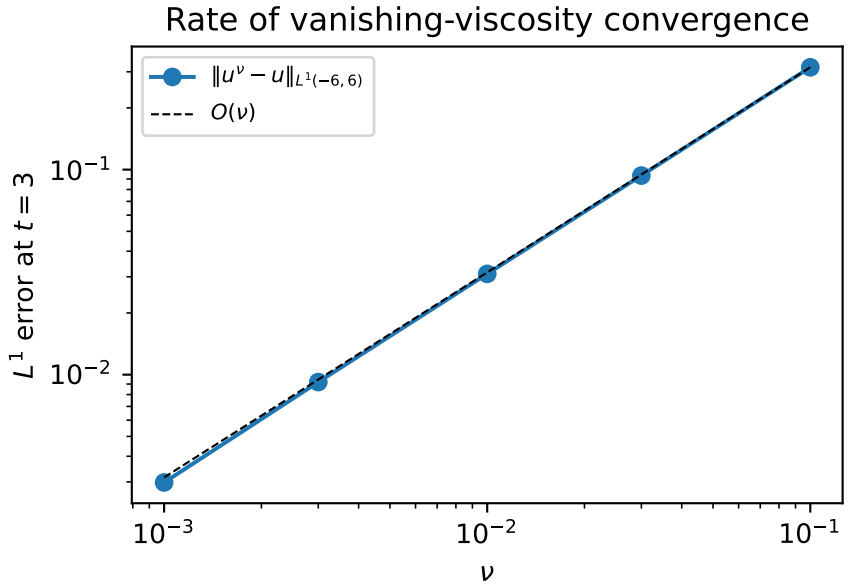

Cole--Hopf solutions as $\nu \to 0$
entropy solution
$\nu = 0.1$
$\nu = 0.03$
$\nu = 0.01$
$\nu = 0.003$
$\nu = 0.001$
$u(x, 3)$
x
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−3
−2
−1
0
1
2
3


Rate of vanishing-viscosity convergence
$\|u^\nu - u\|_{L^1(-6,6)}$
$O(\nu)$
L^1 error at $t = 3$
ν
10^{-1}
10^{-2}
10^{-3}